

AI-Enhanced Task Tracker

Project Description

The AI-Enhanced Task Tracker is a Python-based application that helps users manage tasks efficiently. It allows adding, viewing, editing, and deleting tasks while ensuring a smooth workflow. Additionally, it integrates AI-powered task recommendations using Groq API to optimize task prioritization and productivity.

Features

- Add new tasks with task name, description, priority, due date, and status.
- View tasks in a structured table with sorting options.
- Edit or delete existing tasks.
- Update task status (To-Do, In Progress, Completed).
- Get insights into completed and pending tasks.
- AI-powered task recommendations for better productivity.

Technologies Used

- **Python** (Core Programming Language)
- **Pandas** (Data handling and manipulation)
- **CSV** (Data storage and retrieval)
- **Rich** (Enhanced terminal output and formatting)
- **Groq API** (AI-powered task recommendations)
- **Datetime** (Date handling)

Installation

Prerequisites

Ensure you have Python installed (version 3.7+ recommended). Install the required dependencies:

```
pip install pandas rich groq
```

Setting Up the Project

1. Clone this repository or download the source files.
2. Ensure you have a valid Groq API key and set it as an environment variable: `export GROQ_API_KEY="your_api_key_here"`
3. Run the script: `python script.py`

Usage

1. Main Menu Options:

- a. (1) Add Task
- b. (2) View Tasks (with sorting options)
- c. (3) Edit Task
- d. (4) Delete Task
- e. (5) Update Task Status
- f. (6) Show Insights
- g. (7) AI Recommendations
- h. (8) Exit

2. AI-Powered Recommendations:

- a. Ensure your Groq API key is set up.
- b. AI suggestions help in prioritizing tasks and improving workflow.

Task Insights

- The application provides statistics on completed, pending, and total tasks to track productivity.

Error Handling

- Prevents setting past due dates.
- Validates user inputs for correctness.
- Handles missing CSV files and initializes them automatically.

Future Enhancements

- Add a GUI version using Tkinter or PyQt.
- Integrate with calendar applications.
- Improve AI recommendations with additional context.

License

This project is open-source and free to use.

Author

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